



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology(FST)

Software Engineering

**AIUB Portal Redefined: Integrating Campus Life with
Smart Digital Innovation**

Submitted To

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Project Risk Management

Submitted By

Semester: Summer 2024-2025		Section: D	Group Number: 01	
SN	Student Name	Student ID	Contribution (CO4)	Individual Marks
01	NILOY PAUL	23-51773-2		
02	ARNOB SARKER SUPTA	23-52080-2		
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05	KAHIUM AHAMED FAHIM	22-48593-3		

The project will be Evaluated for the following Course Outcomes

CO4: Develop project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Rubric for Project Assessment (CO4)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Students vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Students fairly provided analysis to the impact of societal, health, safety, legal and cultural issues in their project	Students comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

Project Risk Management

Risk management for the AIUB Portal project entails identifying potential uncertainties, assessing their likelihood and impact, and developing strategies to mitigate them. This ensures successful integration of campus features like authentication, gym registration, cloud printing, lost & found, e-books, faculty directory, anonymous complaints, file sharing, notices, academic registration, and location mapping. Using an agile process, risks are monitored through sprints, with regular reviews to adjust plans, minimizing disruptions to timeline, budget, and quality while addressing societal and environmental sustainability.

Risk Categories:

- PS (Product Size): Risks tied to software scope, requirements, and overall size.
- BU (Business Impact): Risks from management, market, or budget constraints.
- CU (Customer Characteristics): Risks involving user sophistication and communication.
- PR (Process Definition): Risks related to software process adherence and definition.
- DE (Development Environment): Risks in tools, facilities, and development support.
- TE (Technology to be Built): Risks from system complexity and new technologies.
- ST (Staff Size and Experience): Risks due to team skills, experience, and availability.

Impact values:

1. Catastrophic
2. Critical
3. Marginal
4. Negligible

RMMM:

The work product is called Risk Mitigation, Monitoring, and Management Plan. RMMM strategies derived from checklists and Boehm's top risks.

Risk Management Table

Risk	Category	Probability	Impact	RMMM
Feature scope expansion beyond initial estimates	PS	65%	2	Mitigate: Define clear scope in SRS; use agile backlog prioritization. Monitor: Track requirements changes in sprint reviews. Manage: Implement changing control board for approvals.
Unexpected surge in concurrent users during peak times	PS	35%	3	Mitigate: Design scalable architecture with cloud resources. Monitor: Use performance metrics dashboards. Manage: Conduct stress tests and auto-scale setups.

Risk	Category	Probability	Impact	RMMM
Limited reuse of existing campus APIs	PS	55%	2	Mitigate: Assess APIs early via prototypes. Monitor: Review of integration progress weekly. Manage: Develop custom wrappers as backups.
User reluctance to adopt new portal features	BU	45%	3	Mitigate: Run beta testing with feedback loops. Monitor: Track usage analytics post-launch. Manage: Offer incentives and tutorials for adoption.
Shortened project timeline due to academic calendar	BU	60%	2	Mitigate: Build buffers in Gantt chart. Monitor: Weekly progress meetings. Manage: Reprioritize non-core features if needed.
Loss of stakeholder support or funding	CU	30%	1	Mitigate: Regular demos to stakeholders. Monitor: Feedback surveys. Manage: Secure alternative resources early.
Frequent requirement changes from university admins	CU	75%	2	Mitigate: Use formal requirement elicitation. Monitor: Change logs. Manage: Agile iterations for quick adaptations.
New tech like real-time location failing expectations	TE	40%	1	Mitigate: Proof-of-concept trials. Monitor: Tech evaluations. Manage: Fallback to simpler GPS methods.
Insufficient training on Figma or dev tools	DE	70%	3	Mitigate: Schedule workshops. Monitor: Skill assessments. Manage: Pair junior with senior developers.
Team members lacking experience in web security	ST	50%	2	Mitigate: External training sessions. Monitor: Code reviews. Manage: Consult experts for critical modules.
High team turnover mid-project	ST	40%	2	Mitigate: Foster team building. Monitor: Retention check-ins. Manage: Cross-training and documentation.

Risk	Category	Probability	Impact	RMMM
Integration failures between modules like printing and payments	TE	60%	2	Mitigate: Incremental integration testing. Monitor: CI/CD pipelines. Manage: Dedicated integration sprints.
Data privacy breaches in file sharing or complaints	TE	45%	1	Mitigate: Implement GDPR-compliant security. Monitor: Security audits. Manage: Incident response plan
Performance lags in e-book downloads during high demand	PR	50%	3	Mitigate: Optimize with caching. Monitor: Load monitoring tools. Manage: Throttle access if overloaded
Inaccurate faculty directory updates	CU	55%	3	Mitigate: Automated sync with HR systems. Monitor: Data validation checks. Manage: Manual review protocols
Gym slot overbooking due to concurrency issues	PS	30%	2	Mitigate: Use locking mechanisms in database. Monitor: Transaction logs. Manage: Alert system for conflicts.